



Exam 1 Review Problem Set

Differential Equations

Summer 2026

Instructions:

The set of questions serves as PILOT practices to Exam 1 for the Summer 2026 term of AS.110.302 Differential Equations and Applications at Johns Hopkins University.

- Problems marked with an asterisk are more challenging.
- The questions are designed to be done without notes and calculators. Please try your best to work through the problem on your own before looking at the solution.
- Questions on this review problem set are welcomed during PILOT office hours.

1. Solve the following initial value problem (IVP) on $y = y(x)$, and specify the domain for your solution:

$$\begin{cases} y' = (x \log x)^{-1}, \\ y(e) = -6. \end{cases}$$

2. Suppose $f(x)$ is non-zero, let an initial value problem be:

$$\begin{cases} \frac{1-y}{x} \cdot \frac{dy}{dx} = \frac{f(x)}{1+y}, \\ y(0) = 0. \end{cases}$$

- (a) Show that the differential equation is **not** linear.

For the next two questions, suppose $f(x) = \tan x$.

- (b) State, without justification, the open interval(s) in which $f(x)$ is continuous.

- (c)* Show that there exists some $\delta > 0$ such that there exists a unique solution $y(x)$ for $x \in (-\delta, \delta)$.

Now, suppose that $f(x)$ is some function, **not** necessarily continuous.

- (d) Suppose that the condition in (c) does **not** hold, give three examples in which $f(x)$ could be.

3. Solve for the general solution to the following ODEs with $y = y(t)$:

(a) $2y' + y = 3t.$

(b) $y' + \log(t)y = t^{-t}.$

4. Solve the following initial value problem (IVP) on $y = y(x)$, and specify the domain for your solution:

$$\begin{cases} y' = y(y+1), \\ y(0) = 1. \end{cases}$$

5. Given an initial value problem:

$$\begin{cases} \frac{dy}{dt} - \frac{3}{2}y = 3t + 2e^t, \\ y(0) = y_0. \end{cases}$$

- Find the integrating factor $\mu(t)$.
- Solve for the particular solution for the initial value problem.
- Discuss the behavior of the solution as $t \rightarrow \infty$ for different cases of y_0 .

6. An autonomous differential equation is given as follows:

$$\frac{dy}{dt} = 4y^3 - 12y^2 + 9y - 2 \quad \text{where } t \geq 0 \text{ and } y \geq 0.$$

Draw a phase portrait and sketch a few solutions with different initial conditions.

7. Draw the phase line and determine the stability of each equilibrium for the following autonomous differential equations:

- $y' = y^4 - 3y^3 + 2y^2.$
- $y' = y^{2025} - 1.$

8. Let a differential equation be defined as follows:

$$\frac{dy}{dx} = e^{2x} + y - 1.$$

- What is the integrating factor ($\mu(x)$) for the equation? Solve for the general solution.
- Is the equation *exact*? If not, make it exact, then find the general solution.
- Do solutions from part (a) and (b) agree?

- 9.* This brief digression to “differential forms” aims for the following goals:

- Legitimize $\frac{\partial y}{\partial x} = \frac{f(x)}{g(y)} \iff g(y)dy = f(x)dx$ via the differential operator d .
- Get the foundation of *exactness* for certain differential equation relationship.

First, consider variables $x_1, x_2, \dots, x_n$, we may defined the wedge product ($\wedge$) to connect any two variables satisfying that:

$$x_i \wedge x_j = -x_j \wedge x_i \text{ for all } 1 \leq i, j \leq n.$$

- Show that $x_i \wedge x_i = 0$ for $1 \leq i \leq n$.

Now, given any smooth function f , we defined the differential operator (d) as:

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

(b) Suppose $y(x) = e^x$, find dy .

(c) Now, suppose that $\frac{\partial y}{\partial x} = \frac{f(x)}{g(y)}$, can you express dy in terms of the differential form of x .

Note: Since we have just one variable, we have $dy/dx = \partial y/\partial x$, leading to our first goal.

Furthermore, we can apply the differential operator over differential forms with wedge products already. Suppose:

$$\omega = \sum_{i_1, \dots, i_k} f_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

we may have the differential of ω as:

$$d\omega = \sum_{i_1, \dots, i_k} (df_{i_1, \dots, i_k}) dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

(d) Suppose x, y are the variables, and $\omega = 2xy^2 dx + 2x^2 y dy$, show that $d\omega = 0$.

This then relates to a concept called *exactness* in differential equations. Consider the equation:

$$\frac{dy}{dx} + \frac{F(x, y)}{G(x, y)} = 0,$$

we can rewrite it as $F(x, y)dx + G(x, y)dy = 0$. Exactness enforces that:

$$\frac{\partial F}{\partial y} = \frac{\partial G}{\partial x}.$$

Similarly, exactness is considering finding a solution $f(x, y) = c$ such that $F = \frac{\partial f}{\partial x}$ and $G = \frac{\partial f}{\partial y}$.

(e) Show that $df = F(x, y)dx + G(x, y)dy$ and exactness is equivalently $d(df) = 0$.

Note: This implies that the differential equation in part (d) satisfies *exactness*.

10.* Determine if the following differential equation is exact. If not, find the integrating factor to make it exact. Then, solve for its general solution;

$$y'(x) = e^{2x} + y(x) - 1.$$

11. For the first-order autonomous ODE:

$$\frac{dy}{dt} = \sin y + C,$$

where $C \in \mathbb{R}$ is a parameter. Determine any and all bifurcation values for the parameter C and sketch a bifurcation diagram.

12. Let a first order IVP on $y := y(t)$ be defined as follows:

$$\begin{cases} y' = \frac{2}{t}y, \\ y(1) = 1. \end{cases}$$

- (a) Find the solution to the above initial value problem.
- (b) Recall the theorem on existence and uniqueness, as follows:

For an IVP in simple form:

$$\begin{cases} \frac{dy}{dt} = a(t)y + b(t), \\ y(t_0) = y_0. \end{cases}$$

For some $I = (\alpha, \beta) \ni t_0$, if $a(t)$ and $b(t)$ are continuous on the interval I . Then, there exists a unique solution to the IVP on the interval I .

Show that the IVP in this problem does not satisfy the condition for the existence and uniqueness theorem for $\mathbb{R}$.

- (c) Does the above example violates the existence and uniqueness theorem? Why?
13. Let a differential equation be defined as:

$$\frac{dy}{dt} = t - y \text{ and } y(0) = 0.$$

Use Euler's Method with step size $h = 1$ to approximate $y(5)$.

14. For the first-order autonomous ODE:

$$\frac{dx}{dt} = x^2 - 2x + c,$$

with parameter $c \in \mathbb{R}$, do the following:

- (a) Sketch all of the qualitatively different graphs of $f(x) = x^2 - 2x + c$, as c is varied.
 - (b) Determine any and all bifurcation values for the parameter c .
 - (c) Sketch a bifurcation diagram for this ODE.
15. Carbon-14, a radioactive isotope of carbon, is an effective tool in dating the age of organic compounds, as it decays with a relatively long period. Let $Q(t)$ denote the amount of carbon-14 at time t , we suppose that the decay of $Q(t)$ satisfies the following differential equation:

$$\frac{dQ}{dt} = -\lambda Q \text{ where } \lambda \text{ is the rate of decay constant.}$$

- (a) Let the half-life of carbon-14 be τ , find the rate of decay, λ .
 - (b) Suppose that a piece of remain is discovered to have 10% of the original amount of carbon-14, find the age of the remain in terms of τ .
- 16.* Let an initial value problem be defined as follows:

$$\begin{cases} (12x^4 + 5x^2 + 6) \frac{dy}{dx} - (x^2 \sin(x) + x^3)y = 0, \\ y(0) = 1. \end{cases}$$

Show that the solution to the above initial value problem is symmetric about $x = 0$.

17. Given a differential equation for $y = y(t)$ being:

$$t^3 y'' + t y' - y = 0.$$

- (a) Verify that $y_1(t) = t$ is a solution to the differential equation.
- (b) Find the full set of solutions using reduction of order.
- (c) Show that the set of solutions from part (b) is linearly independent.

18. Solve the following second order differential equations for $y = y(x)$:

- (a) $y'' + y' - 132y = 0.$
- (b) $y'' - 4y' = -4y.$
- (c) $y'' - 2y' + 3y = 0.$

19.* Given the following second order initial value problem:

$$\begin{cases} \frac{d^2 y}{dx^2} + \cos(1-x)y = x^2 - 2x + 1, \\ y(1) = 1, \\ \frac{dy}{dx}(1) = 0. \end{cases}$$

Prove that the solution $y(x)$ is symmetric about $x = 1$, i.e., satisfying that $y(x) = y(2-x)$.

Hint: Consider the interval in which the solution is unique.